

## hovid Lincomycin Injection 600 mg / 2mL

VILINZC-M2

### DESCRIPTION

A clear, colourless or almost colourless solution filled in transparent glass ampoule.

### COMPOSITION

**Each 2mL ampoule contains:** Lincomycin Hydrochloride equivalent to Lincomycin 600 mg.

### ACTIONS AND PHARMACOLOGY

Lincomycin is a lincosamide antibacterial with mainly bacteriostatic action against Gram-positive aerobes and many anaerobic bacteria.

Lincosamides such as lincomycin bind to the 50S subunit of the bacterial ribosome, similarly to macrolides such as erythromycin, and inhibit the early stages of protein synthesis. The action of lincomycin is mainly bacteriostatic, although high concentrations may be slowly bactericidal against sensitive strains.

Lincomycin is active against enterococci, most aerobic Gram-positive and most aerobic Gram-negative bacteria. Susceptible Gram-positive anaerobes include *Eubacterium*, *Propionibacterium*, *Peptococcus*, and *Pepostreptococcus* spp., and many strains of *Clostridium perfringens* and *Cl. tetani*. Gram-negative anaerobes susceptible to lincomycin include *Fusobacterium* spp. (although *F. varium* is usually resistant), *Prevotella* spp., and *Bacteroides* spp., including the *B. fragilis* group.

Cross resistance has been demonstrated between clindamycin and lincomycin.

### PHARMACOKINETICS

An intramuscular injection of 600 mg produces average peak plasma concentrations of between 11 and 12 micrograms/mL at 60 minutes and a 2-hour intravenous infusion of 600 mg produces an average of about 16 micrograms/mL.

The biological half-life after intramuscular or intravenous administration is about 5 hours. The serum half-life of lincomycin may be prolonged in patients with severe impairment of renal function compared to patients with normal renal function. In patients with abnormal hepatic function, serum half-life may be twofold longer than in patients with normal hepatic function. Haemodialysis and peritoneal dialysis are not effective in removing lincomycin from the serum.

Lincomycin is widely distributed in the tissues including bone and body fluids but diffusion into CSF is poor, although it may be slightly better when the meninges are inflamed. It diffuses across the placenta and is distributed into breast milk. Lincomycin is partially inactivated in the liver; unchanged drug and metabolites are excreted in the urine, bile, and faeces.

### INDICATION

Lincomycin HCl Injection is indicated in the treatment of serious infections due to susceptible strains of streptococci, pneumococci, and staphylococci. Its use should be reserved for penicillin-allergic patients or other patients for whom, in the judgment of the physician, a penicillin is inappropriate. Because of the risk of colitis, as described in the WARNING, before selecting Lincomycin the physician should consider the nature of the infection and the suitability of less toxic alternatives (e.g., erythromycin). Lincomycin has been demonstrated to be effective in the treatment of staphylococcal infections resistant to other antibiotics and susceptible to Lincomycin. Staphylococcal strains resistant to Lincomycin have been recovered; culture and susceptibility studies should be done in conjunction with therapy with Lincomycin. In the case of macrolides, partial but not complete cross resistance may occur. The drug may be administered concomitantly with other antimicrobial agents when indicated.

### CONTRAINDICATIONS

This medication should not be used by patients with known hypersensitivity to lincomycin or clindamycin. Lincomycin is contraindicated in newborns.

### WARNINGS AND PRECAUTIONS

- Clostridium difficile* associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents, including Lincomycin, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

*C. difficile* produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibacterial use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

If CDAD is suspected or confirmed, ongoing antibacterial use not directed against *C. difficile* may need to be discontinued. Appropriate fluid and electrolyte management, protein supplementation, antibacterial treatment of *C. difficile*, and surgical evaluation should be instituted as clinically indicated.

- Serious hypersensitivity reactions, including anaphylaxis and erythema multiforme, have been reported with use of Lincomycin. If an allergic reaction to Lincomycin occurs, discontinue the drug.
- Although lincomycin appears to diffuse into cerebrospinal fluid, levels of lincomycin in the CSF may be inadequate for the treatment of meningitis.
- Review of experience to date suggests that a subgroup of older patients with associated severe illness may tolerate diarrhoea less well. When Lincomycin is indicated in these patients, they should be carefully monitored for change in bowel frequency.
- Lincomycin should be prescribed with caution in individuals with a history of gastrointestinal disease, particularly colitis.
- Lincomycin should be used with caution in patients with a history of asthma or significant allergies.
- Certain infections may require incision and drainage or other indicated surgical procedures in addition to antibacterial therapy.
- The use of Lincomycin may result in overgrowth of nonsusceptible organisms-particularly yeasts. Should superinfections occur, appropriate measures should be taken as indicated by the clinical situation. When patients with pre-existing monilial infections require therapy with Lincomycin, concomitant antimicrobial treatment should be given.
- The serum half-life of lincomycin may be prolonged in patients with severe impairment of renal function compared to patients with normal renal function. In patients with abnormal hepatic function, serum half-life may be twofold longer than in patients with normal hepatic function.
- Patients with severe impairment of renal function and/or abnormal hepatic function should be dosed with caution and serum lincomycin levels monitored during high-dose therapy.
- Lincomycin should not be injected intravenously undiluted as a bolus, but should be infused over at least 60 minutes.
- Prescribing Lincomycin in the absence of a proven or strongly suspected bacterial infection or a prophylactic indication is unlikely to provide benefit to the patient and increases the risk of the development of drug-resistant bacteria.
- Lincomycin therapy has been associated with severe colitis which may end fatally.

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- It should be reserved for serious infections where less toxic antimicrobial agents are inappropriate.
- It should not be used in patients with nonbacterial infections, such as most upper respiratory tract infections.

**PREGNANCY AND LACTATION****Pregnancy**

There have been no adequate data; it is recommended that Lincomycin not be used in pregnant women.

**Lactation**

Lincomycin has been reported to appear in human milk in concentrations of 0.5 to 2.4 mcg/mL. Because of the potential for serious adverse reactions in nursing infants from lincomycin, a decision should be made whether to discontinue nursing, or to discontinue the drug, taking into account the importance of the drug to the mother.

**DRUG INTERACTIONS**

- *In vitro* studies have shown antagonistic activity between Lincomycin and erythromycin; therefore, these agents should not be used concurrently.
- Lincomycin has neuromuscular blocking properties which may enhance the action of other neuromuscular blocking agents. It should be used with caution in patients receiving such agents.
- Absorption of lincomycin is reduced by adsorbent antidiarrheal and cyclamate sweeteners.
- hovid Lincomycin Injection is physically incompatible with novobiocin, kamarycin and phenytoin.

**MAIN SIDE/ ADVERSE EFFECTS**

The following reactions have been reported with the use of lincomycin:

**Gastrointestinal**

Glossitis, stomatitis, nausea, vomiting, antibacterial-associated diarrhea and colitis, and pruritus ani. Onset of pseudomembranous colitis symptoms may occur during or after antibacterial treatment.

**Hematopoietic**

Neutropenia, leukopenia, agranulocytosis and thrombocytopenic purpura have been reported. There have been rare reports of aplastic anaemia and pancytopenia in which Lincomycin could not be ruled out as the causative agent.

**Hypersensitivity Reactions**

Hypersensitivity reactions such as angioneurotic oedema, serum sickness and anaphylaxis have been reported. Cases of erythema multiforme, some resembling Stevens-Johnson syndrome, have been associated with Lincomycin.

**Skin and Mucous Membranes**

Skin rashes, urticaria and vaginitis and rare instances of exfoliative and vesiculobullous dermatitis have been reported.

**Liver**

Although no direct relationship of Lincomycin to liver dysfunction has been established, jaundice and abnormal liver function tests (particularly elevations of serum transaminase) have been observed.

**Renal**

Although no direct relationship of lincomycin to renal damage has been established, renal dysfunction as evidenced by azotemia, oliguria, and/or proteinuria has been observed in rare instances.

**Cardiovascular**

After too rapid intravenous administration, rare instances of cardiopulmonary arrest and hypotension have been reported.

**Special Senses**

Tinnitus and vertigo have been reported occasionally.

**Local Reactions**

Patients have demonstrated excellent local tolerance to intramuscularly administered Lincomycin. Reports of pain following injection have been infrequent. Intravenous administration of Lincomycin in 250 to 500 mL of 5% dextrose injection or 0.9% sodium chloride injection produced no local irritation or phlebitis.

**OVERDOSE AND TREATMENT**

Serum levels of lincomycin are not appreciably affected by haemodialysis and peritoneal dialysis.

**DOSAGE AND ADMINISTRATION****Intramuscular**

*Adults:* 300 mg - 600 mg of Lincomycin intramuscularly every 12 hours.

*Children over 1 month of age:* One or two intramuscular injection of 10 mg/kg every 24 hours.

**Intravenous**

*Adults:* The intravenous dose will be determined by the severity of the infection.

Two or three intravenous infusion of 600 mg every 24 hours.

*Children over 1 month of age:* Two or three intravenous infusion of 10-20 mg/kg every 24 hours.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

**Storage:**

Store below 25°C.

Protect from light and freezing.

**Presentation/Packing:**

10 ampoules of 2 mL

**Product Registration Holder/Repacked by: HOVID Bhd.**

121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.

Manufactured by: NCPC Hebei Huamin Pharmaceutical Co., Ltd.,  
No. 98, Hainan Road, Economic & Technological Development Zone,  
Shijiazhuang, China.

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